Predicting social effects on pharmaceutical wastewater concentrations

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ABSTRACT

Pharmaceuticals in the environment cause myriad problems including toxicity in aquatic organisms, proliferation of antimicrobial resistance, and water reuse challenges. Predicting pharmaceutical occurrence in wastewater, a primary source of pharmaceuticals in the environment, may inform pharmaceutical waste management. Previous models have predicted pharmaceutical concentration from consumption data, but few models have explored the impact of demographics on a given community's pharmaceutical consumption. This study trains a suite of machine learning models on pharmaceutical consumption and demographics data to predict a given community's wastewater pharmaceutical concentration profile. Of the six models implemented, the multi-output k-nearest neighbor model produced the most accurate predictions for drug name (recall: 0.99), and the multi-output decision tree model produced the most accurate predictions for concentration (recall: 0.87). Most models were not sensitive to feature reduction, with the notable exception of the random forest model, whose accuracy increased 34 to 36% after removing each feature except those related to socioeconomic status, suggesting possible importance of these variables. Exploratory data analysis also demonstrated that pharmaceutical concentrations, including antibiotic and opioid concentrations, generally decreased between 2014 and 2022, and that the mass of select cold and flu drugs did not increase much at the onset of the COVID-19 pandemic. Improved hyperparameter fitting, especially for the socioeconomic status variables, is necessary to produce more robust conclusions and trustworthy models. Furthermore, while the crises of antimicrobial resistance and the opioid epidemic appear to be driving a net decrease in use of these drugs, additional temporal analysis is needed.

KEYWORDS
pharmaceuticals, wastewater influent, demographics, machine learning
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1 INTRODUCTION

1.1 Motivation

In 2009, Rockström et al. proposed the concept of planetary boundaries, or quantitative thresholds for key planetary processes (e.g., climate change) beyond which human development is no longer sustainable [1]. If these boundaries are transgressed, then Earth may become destabilized and enter a new geological era (the Anthropocene), which would be unprecedented in human history and may challenge humans' survival. As of 2023, the thresholds for 9 planetary boundaries have been quantified: climate change, biosphere integrity, land system change, freshwater change, biogeochemical flows, ocean acidification, atmospheric aerosol loading, stratospheric ozone depletion, and novel entities [2]. According to this most recent model, 6 of the 9 planetary boundaries have been transgressed (Figure 1).

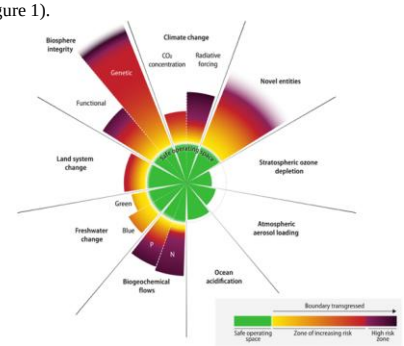

Figure 1: The 2023 planetary boundary model [2].

To mitigate the potential shift into the Anthropocene era, the planetary boundary model needs to be further refined, and actionable steps need to be identified. This research will contribute toward these goals for the novel entities boundary, which is among the six transgressed boundaries. Novel entities include novel chemicals, synthetic organisms, and materials or life forms altered or mobilized by human activity [3]. This study will focus on pharmaceuticals, one class of novel entities.

Novel entities typically undergo rapid development, which challenges their regulation. Pharmaceuticals are no exception; on average, the U.S. Food & Drug Administration (FDA) approves a new pharmaceutical within 8.7 years of initial investigation (Figure 2) [4].

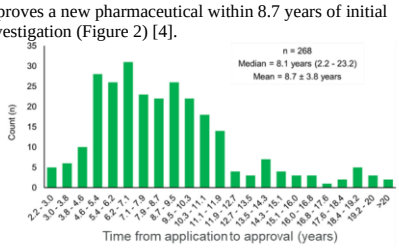

Figure 2: Distribution of FDA approval time for pharmaceuticals in the U.S. from 2010-2019, adapted from [4].

Rapid pharmaceutical development enables beneficial health outcomes such as treatment for new or previously untreatable diseases or improved management of medical conditions. However, pharmaceuticals in the environment are contaminants, and their development outpaces the ability of regulatory agencies to manage them. In fact, regulation of chemical contaminants under the U.S. Safe Drinking Water Act often takes 25 years or longer. For example, the first lawsuit to regulate carcinogenic per- and poly-fluoroalkyl substances (PFAS), also known as forever chemicals, was initiated in 1998, but PFAS were not regulated in drinking water until April 2024. [5, 6]. This discrepancy suggests that non-regulatory frameworks such as collaboration between researchers and wastewater utilities will be needed in addition to regulations to manage pharmaceutical pollution.
Pharmaceuticals primarily enter the environment through residential wastewater, emissions from pharmaceutical manufacturing facilities, and agricultural runoff, mostly affecting organisms living in impacted surface waters and riparian organisms within the same food web. In these non-target organisms, pharmaceuticals have demonstrated ecotoxicological effects such as increased cancer rates and physiological and reproductive dysfunction [7, 8, 9]. In addition to the broad ecotoxicity of pharmaceuticals, releasing antimicrobial drugs into the environment drives antimicrobial resistance, which has been identified as one of the greatest public health challenges of the next century [7, 10, 11].

If advanced wastewater treatment technology is available, most pharmaceuticals should be sufficiently removed from wastewater [12]. However, many small and low-income communities are not served by plants with advanced treatment capability, especially in low-income countries [12, 13, 14, 15]. This means that the people living in these communities as well as the ecosystems associated with them are experiencing pharmaceutical pollution. If the community uses surface water as a drinking water source, then residents' drinking water may be contaminated with a complex mixture of pharmaceuticals. This effect is pronounced in dry regions of the world, where wastewater discharges may comprise a significant portion of the flow of local streams and rivers [16, 17]. While these pharmaceuticals are present at low levels, they are designed to be bioactive at such levels and therefore may cause physiological effects [9]. Furthermore, the mixture effects of pharmaceuticals on human health have not been well-studied and may lead to adverse health outcomes [18]. Pharmaceuticals also challenge water reuse, which is being explored as a potential solution to climate-driven water scarcity [12, 16]. Because pharmaceuticals are resistant to degradation and may become more concentrated during reuse if they are not sufficiently removed, pharmaceuticals make water reuse more challenging and expensive to implement [9, 12].

1.2 Background

Pharmaceuticals in wastewater have been widely studied, with over 310,000 publications in the literature since 1975 (see Figure 3) [19].

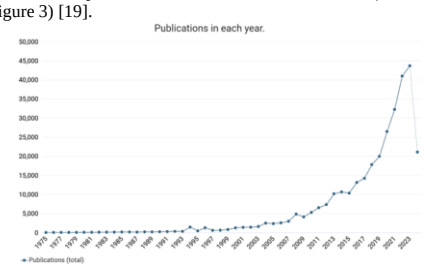

Figure 3: Number of publications on pharmaceuticals in wastewater per year, 1975-2024 [19].

One approach to pharmaceutical management is to predict expected pharmaceutical concentrations entering wastewater treatment facilities, which facilitates optimized wastewater treatment strategies without the need for regular monitoring. Such models typically use aggregated pharmaceutical sales or consumption data to predict the expected mass of each pharmaceutical entering a wastewater treatment facility [20, 21, 22]. Model output can be used to identify pharmaceuticals likely to pass through wastewater treatment, resist degradation, or exceed toxicity levels, which can inform treatment prioritization. They can also assist in environmental justice efforts by locating sites that might require additional monitoring for pharmaceutical contamination without initially requiring expensive chemical analysis. Finally, they can be used to determine what treatment strategies are most successful at removing different pharmaceuticals, potentially informing more effective treatment trains. Despite these benefits, models to predict pharmaceutical occurrence in the United States remain limited, especially within the past decade. While there are numerous and recent models available for other countries, pharmaceutical regulations, access, and consumption patterns vary widely between countries [9, 23].

Beyond consumption, pharmaceutical wastewater concentrations can be influenced by many factors including prescription cost

and insurance coverage, demographics, over-the-counter and other non-prescription sources, and chemical reactions and partitioning processes (e.g., sorbing to sewer sediments) in the sewer system [22, 24, 25]. The goal of this research is to develop a model that accounts for the impact of social characteristics (e.g., demographics) on pharmaceutical wastewater concentrations.

Previously, Maybruck and Mansfeldt developed a suite of tools called SeweRx. SeweRx has two modules: PharmUse and PharmFlush (Figure 4) [26].

Figure 4: SeweRx, PharmUse, and PharmFlush logos.

PharmUse is a database that synthesizes information on 290 pharmaceuticals from the 2020 U.S. Agency for Healthcare Research & Quality Medical Expenditure Panel Survey (MEPS) and the U.S. Environmental Protection Agency CompTox Chemicals Dashboard v2.2.0 [27, 28]. Data includes daily mass load per pharmaceutical, number of prescriptions per pharmaceutical, duration of prescription, and the identifiers (SMILES, InChI string, InChIKey), physicochemical properties, Lethal Concentration 50 (LC50; concentration that kills 50% of test subjects), and No Observed Effect Concentration (NOEC; highest concentration at which no adverse health outcomes occur). PharmFlush is a model trained on PharmUse that employs a binomial distribution model to compute the expected daily wastewater influent concentration of each pharmaceutical in a sewershed (i.e., community served by a wastewater treatment plant) of a given size on any given day.

To evaluate PharmFlush's performance, a systematic literature search was conducted. The literature search included reported pharmaceutical concentration measurements in wastewater influent in the United States in literature published between 2019 and 2024. The resulting dataset is derived from 36 papers (selected from 399 papers) with 2250 recorded measurements and coverage for over 90 pharmaceuticals. All PharmFlush predictions were within an order of magnitude of the literature-reported values.

1.3 Objectives

This research extends PharmFlush and investigates the impact of social factors on pharmaceutical wastewater concentrations. Broadly, this study attempts to predict the pharmaceutical wastewater concentration profile of a given community from its demographic features. The impact of temporal trends on pharmaceutical concentrations are also briefly considered. To this end, the research questions are:

1) How does population age structure affect pharmaceutical influent concentrations?
2) How does gender affect pharmaceutical influent concentrations?
3) How do race and ethnicity affect pharmaceutical influent concentrations?
4) How does income level affect pharmaceutical influent concentrations?
5) How does insurance coverage affect pharmaceutical influent concentrations?
6) How have prescriptions for different pharmaceuticals changed over time between 2014 and 2022?
7) How have antimicrobial prescriptions changed between 2014 and 2022?
8) How have opioid prescriptions changed between 2014 and 2022?
9) How did prescription pharmaceutical usage change during the COVID-19 pandemic?
10) How might social and scientific shifts in response to events like antimicrobial resistance, opioid addiction, and pandemics influence pharmaceutical concentration predictions?

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2 DATA COLLECTION AND PROCESSING

2.1 Data Collection

Data for this study are from the MEPS Prescribed Medicine and Full Year Population Characteristics surveys for years 2014, 2016, 2018, and 2020-2022, U.S. Marshals Service "Over-the-Counter" Medication List, the U.S. Drug Enforcement Administration (DEA) Controlled Substances listing, and the U.S. Food & Drug Administration (FDA) Animal Drugs database, specifically the Green Book, Section 2: Active Ingredients [29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41]. Respectively, these sources provide insight into temporal prescription consumption trends, demographics, over-the-counter (OTC) availability of prescription pharmaceuticals, prescription pharmaceuticals with abuse potential, and prescription pharmaceuticals used in veterinary medicine. Each of these may influence the wastewater influent concentrations at municipal wastewater treatment facilities. Although this project focuses on demographic features and their ability to predict pharmaceutical concentrations, the other variables will be useful in later analyses and therefore are still included.

The MEPS data were all downloaded directly from the Agency for Healthcare Research & Quality website (links provided in the Bibliography). For years 2014 and 2016, data were provided only in ASCII and SAS format. The SAS files were converted to csv files for analysis using the read_xpt() function in R's haven library. Years 2018 and 2020-2022 were available to download as Excel workbooks. The MEPS Prescribed Medicine and Population Characteristics surveys were integrated by an ID column that uniquely identifies each person across MEPS datasets. The OTC list was available for download from the U.S. Marshals Service as a PDF file. The list of interest was extracted using Python's pdfplumber package. The Controlled Substances listing was available for download as a PDF file and consisted solely of the table of controlled substances; it was converted to an Excel workbook using Adobe Acrobat's built-in functionality. Finally, the Active Ingredients file was available

for direct download as an Excel workbook from the FDA Animal Drugs database website.

Together, these data provide a rich source of information. The MEPS Prescribed Medicine surveys each contain between 200,000 and 300,000 rows and 65-66 columns of data each prior to cleaning, while the MEPS Population Characteristics surveys each contain about 30,000 rows and 1,840 columns. After cleaning, the Prescribed Medicine datasets still include over 200,000 rows each, and the Population Characteristics datasets still include over 30,000 rows each. This demonstrates reasonably high data quality after removing noise and handling missing values. The other datasets are of sufficient quality for inclusion as well. The OTC list includes 188 pharmaceuticals, including the most common OTC pharmaceuticals. The controlled substances list is the official guidebook for the DEA, has been updated as recently as July, and includes 579 controlled substances including opioids (clearly labeled by the presence of a binary Narcotics column). Finally, the Active Ingredients file from the Green Book contains all drugs approved for veterinary use by the FDA as of January (2236 drugs total).

2.2 Data Cleaning and Pre-processing

Since all their data is household-reported and is inconsistent between years, the MEPS Prescribed Medicines datasets required extensive pre-processing via the following steps:

1) Data reduction was performed, eliminating columns that do not bear relevance to the 10 questions identified in Objectives. Columns related to calculating the daily dosage for each prescription, identifiers for the person and pharmaceutical, and cost data for the prescription were retained.
2) The columns were renamed to be clearer than their abbreviations.
3) Numeric variables were converted to the numeric datatype.
4) Missing values, represented as negative numbers in the data, were replaced by NA. If >95% of the strength, day supply, or quantity variables were NA for a given drug, that drug was removed from consideration for further analysis. If <95% of the values for these variables were NA, then the mean of the available values was calculated, and the mean was used to impute the NA values.
5) Drug administration route (nasal, topical, otic, ophthalmic) is provided together with the drug name column in the original data. The administration route now has its own column, Administration_route, which can be used together with the Form column for analysis.
6) The values of the Form variable are given only as abbreviations, which are not always clear. The abbreviations are replaced by the full word for each Form for interpretable analysis. If the Form variable coding is not provided, then it is left as NA.
7) Rows with unclear units (e.g. “Other,” “Unit/mL”) are removed from the dataset. Rows with extraneous information in the Units were corrected (e.g., mg/blister was corrected to mg).
8) Unit conversions were performed to get all the strength values in micrograms.
9) Combination drugs (e.g., amoxicillin-clavulanate) were separated. These drugs have dashes or slashes separating values in the Drug, Strength, and Units columns. The Units column was handled carefully, since some units should have slashes (e.g., mg/mL).
10) Prescriptions for substances found in food or naturally occurring in the body are not distinguishable from these other sources in wastewater. For example, prescriptions for vitamin K should be excluded since prescribed vitamin K cannot be distinguished in a wastewater sample from food-derived vitamin K. Each dataset is checked against a list of commonly prescribed substances that should be excluded from the analysis and against the list of pharmaceuticals from the 2020 analysis. If the given dataset contains more substances to be excluded, those substances are added to the list of substances to exclude moving forward.

After each pharmaceutical is processed for each MEPS survey, five new columns are added: Daily_Frequency (Quantity/Day_Supply), Daily_Dosage (Strength*Daily_Frequency), Year (useful after integration with other MEPS data), Concentration, and Concentration_Level. Concentration was determined by dividing by the volume of wastewater flowing through the sewershed (assuming 100,000 people in the sewershed and the U.S. average of 310.404 L of wastewater produced per person). The wastewater volume was also multiplied by a correction factor of 4 to account for medication non-compliance (estimated at 50%) and for the effect of employees coming into the sewershed for work [42]. Concentration_Level was determined by calculating thresholds at the quantiles of the Concentration variable, which resulted in the Low, Medium, High labels used in the dataset. The Population Characteristics files are more straightforward with the same process carried out for each year. After selecting the desired columns of or related to age, sex, race, ethnicity, income, and insurance coverage, missing values are imputed by mean for age and prescription insurance coverage (which was numerically encoded and averaged between different rounds of the survey) and by mode for ability to obtain prescription medication.

After processing the individual datasets, the population data were joined with the prescription data by using R’s left_join() function on the Person ID column. Then all the MEPS data were joined using the rbind() function. The binary variables OTC_drug, Abuse_potential, and Veterinary_drug were created to indicate if a given drug falls into any of these categories, and then the OTC, controlled susbtances, and veterinary data were similarly integrated with the other data using left_join(). This data cleaning process is displayed in Figure 5.

a)

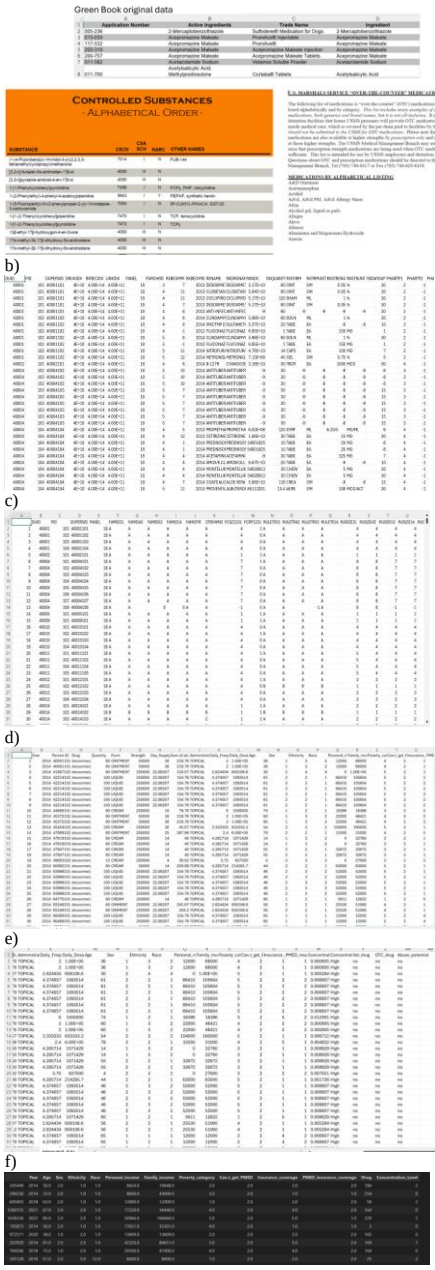


Figure 5. a) Original data for MEPS Prescribed Medicines file during 2014. b) Original data for MEPS Population Characteristics file during 2014. c) Original data for the OTC, controlled substances, and veterinary pharmaceutical lists. d) Integrated dataset combining Figures 5a-c for all six years, part 1. e) Integrated dataset combining Figures 5a-c for all six years, part 2. f) Integrated dataset following cleaning, pre-processing, label encoding, and definition of the concentration level variable, used in the multi-output decision tree model.

3 MODEL IMPLEMENTATION AND RESULTS

3.1 Exploratory Data Analysis:

The data contains both numeric and categorical attributes. Numeric attributes are all ratio-scaled and include: Daily Dosage, Quantity, Strength, Day Supply, Amount paid self or family, Amount paid Medicare, Amount paid Medicaid, Amount paid private insurance, Amount paid veterans CHAMPVA, Amount paid Tricare, Amount paid other federal, Amount paid state or local, Amount paid workers comp, Amount paid other insurance, Sum of all payments, Daily Frequency, Age, Personal income, Family income, and Concentration. Categorical attributes are either nominal (Person ID, Drug, Form, Administration route, Ethnicity, Race), ordinal (Poverty category, Year, Concentration Level), or binary (Sex, Can't get PMED, Insurance coverage, PMED insurance coverage, Abuse potential, Vet drug, OTC drug). Notably, PMED refers to prescription medication.

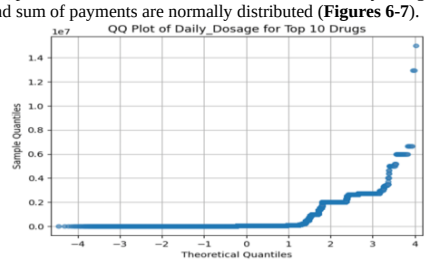
Central tendency and dispersion are characterized for the Daily Dosage and Sum of all payment's attributes (**Table 1**). Daily Dosage is the average dosage of each drug that patients receive, and Sum of all payments is the total cost of each drug.

Table 1. Central tendency and dispersion of Daily Dosage and Sum of all payment's variables.

		Daily dosage	Sum of all payments
Central Tendency	**Mean**	523065.5485	55.2704
	Median	25000.0	10.5
	Mode	20000.0	10.0
Dispersion	**Standard Deviation**	34308161.7922	255.9124
	Variance	1177049965557707.8	65491.1455
	Minimum	0.0006	0.0
	Maximum	22661607263.8569	21640.14
	Range	22661607263.8563	21640.14
	25th Percentile (Q1)	10000.0	5.0
	75th Percentile (Q3)	80000.0	23.54

	Interquartile Range (IQR)	70000.0	18.54

QQ plots are also created to visualize whether the daily dosage and sum of payments are normally distributed (**Figures 6-7**).



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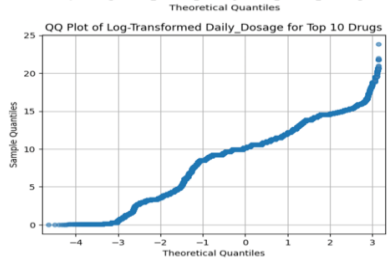
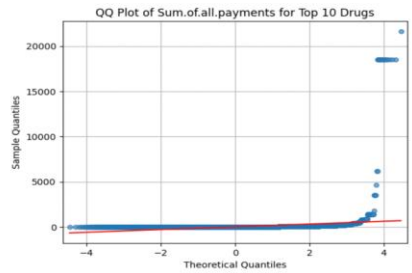


b)

Figure 6. a) QQ plot for daily dosage for 10 drugs with highest daily dosage, b) Log-transformed QQ plot.

For daily dosage, there are significant deviations of the data points from a normal distribution (**Figure 6a**). Many drugs have daily dosages at around 0.0, and there is an upward curve at the end, which indicates that there are more extreme values than in a normal distribution. The plot also has a right-skewed distribution with most of the values clustered at the lower end with a few extreme values. After applying a log transformation, the overall normality of the data improved (**Figure 6b**). For sum of payments, the lower and the middle part (i.e. theoretical quantiles -4 to 2) of the distribution closely follows the normal distribution (**Figure 7a**). The plot has an upward curve after theoretical quantile 2, which indicates a heavier upper tail compared to a normal distribution. This means there are some unusually high payment values and those are causing skewness in the distribution. The Sum of Payments data is right-skewed, (i.e., most values are lower with a few extreme high values). The sample quantile is around 0 for the lower quantiles, suggesting that there are a lot of payments with values close to 0. Applying a log transformation makes the data more normally distributed (**Figure 7b**).



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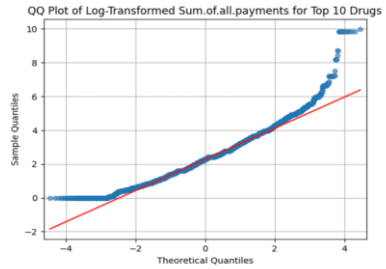
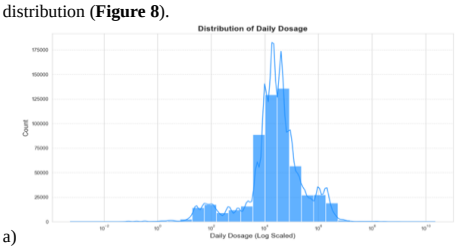


b)

Figure 7. a) QQ plot for sum of payment for 10 drugs with highest daily dosage. b) Log-transformed QQ plot.

Histograms were also generated for these variables to see their distribution (**Figure 8**).



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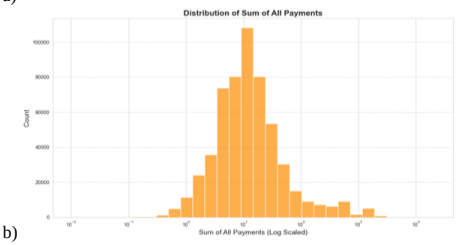


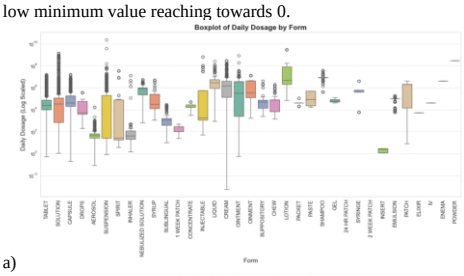
b)

Figure 8. a) Distribution of daily dosage, and b) distribution of sum of payment.

For daily dosage (**Figure 8a**), the distribution is negatively skewed, with a peak at around 10^5, and tails off to the left, suggesting that, while smaller doses are possible, they are quite rare. The distribution is also multimodal, likely reflecting different typical dosages for varying medical needs or conditions. For sum of payments (**Figure 8b**), the histogram

resembles a bell curve, indicating that the data is approximately log-normally distributed. The peak of the distribution lies around 10^1, indicating that the majority of the sums fall within this range. The right tail suggests a positive skew; there are some larger payment sums, though they occur less frequently.

Categorical variables were visualized as they relate to daily dosage and sum of payments. Different forms of pharmaceuticals had different dispersion of the daily dosage variable (**Figure 9a**). Tablets, solutions, and suspensions show high variability in daily dosages, with many outliers and a broad range of values, likely due to their use in diverse medical cases with occasional high-dosage needs. This may indicate occasional very high dosages in special cases. Capsules, aerosols, creams, and ointments also contain many outliers with a smaller but still significant range of values. Cream has a very low minimum value reaching towards 0.



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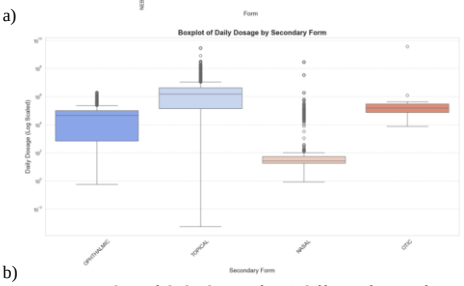


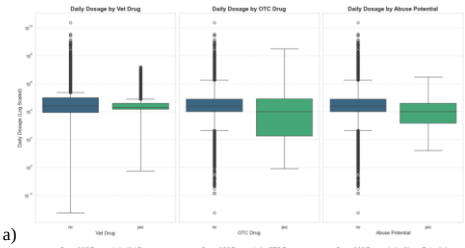
b)

Figure 9. Boxplots of daily dosage for a) different forms of pharmaceutical and b) different administration routes (or secondary forms as defined in this paper).

Daily dosage also experiences dispersion for different administration routes (also called secondary forms in this paper) (**Figure 9b**). For ophthalmic, there is a large interquartile range (IQR) extending from 10^3 to 10^5, with a median of about 10^4, and there are several outliers above the upper whisker until around 10^7. For topical, the median is relatively high at around 10^7, the IQR is from around 10^5 to 10^7, and there are several outliers above the whisker reaching to around 10^9. There is also a very low minimum at around 10^{-3}. For nasal, the median is relatively low at around 10^1, the IQR is small (between 10^1 to 10^2), and the outliers range greatly with the highest reaching 10^9. For otic, the median lies at around 10^5, the IQR is relatively small, ranging from around 10^5 to 10^6, and there are very few outliers with the highest reaching 10^{10}. Overall, topical and otic drugs have the highest average daily dosages, and nasal drugs have the lowest daily dosages (with some high-dosage outliers).

Further, there are very few outliers for daily dosage for otic drugs, and ophthalmic drugs have a large IQR, which suggests a high variability in typical dosages.

There are also differences in dispersion in daily dosage and sum of payment for the veterinary drug, OTC drug, and abuse potential categories (**Figure 10**).



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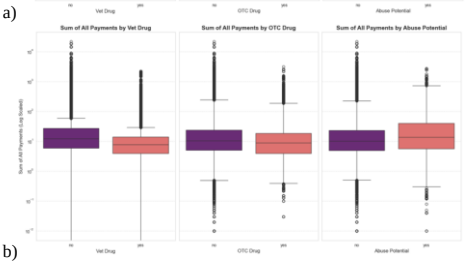


b)

Figure 10. Boxplots of veterinary drug use, over the counter (OTC) drug use, and abuse potential by a) daily dosage, and b) sum of payment.

Specifically, non-veterinary drugs have a slightly higher median daily dosage and a broader range, including more extreme outliers, while veterinary drugs show a narrower dosage distribution and generally lower dosage ranges, which may indicate more standardized dosing practices for veterinary applications (**Figure 10a**). Further, non-OTC drugs tend to have a higher median daily dosage than OTC drugs. This indicates that prescription medications generally require stronger dosing. Additionally, the range for OTC drugs is much broader than non-OTC drugs, suggesting that although the typical dosage is lower than for non-OTC drugs, there is considerable variation in how OTC drugs are used. Non-OTC drugs also exhibit a large number of outliers on both the high and low end, reflecting where dosages need to be tailored more specifically for optimized patient care. In contrast, OTC drugs do not have any outliers, reflecting the more consistent and controlled dosing patterns. Finally, drugs without abuse potential tend to have a slightly higher median dosage than drugs with abuse potential. The distribution for drugs without abuse potential shows a larger number of outliers, indicating high variability in daily dosage. The IQR for drugs with abuse potential is a bit broader than drugs without abuse potential, but drugs with abuse potential have no outliers for daily dosage, suggesting tighter control of dosing to limit abuse or overdose risk.

For sum of payment (**Figure 10b**), the IQR is larger for non-veterinary drugs than veterinary drugs. Both categories have a

large number of outliers above the IQR, especially for non-veterinary drugs, suggesting certain non-veterinary drugs are more expensive. The median line appears to be slightly lower for veterinary drugs compared to non-veterinary drugs. This indicates that veterinary drugs are slightly less expensive than non-veterinary drugs. Further, the IQR for payment cost seems to be about the same for both OTC drugs and non-OTC drugs. There are more extreme outliers in the non-OTC drug category, indicating that non-OTC drugs vary more in price compared to OTC drugs. The medians are also similar with non-OTC drugs being slightly more expensive than OTC drugs. This suggests that on average non-OTC drugs are slightly more expensive. Finally, payments for drugs with abuse potential shows a larger spread and a slightly higher median than those without abuse potential. Drugs with abuse potential are generally more expensive and have a broader range of payment amounts. There are also more outliers for drugs without abuse potential than drugs with abuse potential. This indicates greater variability in payment amounts for drugs without abuse potential, even though drugs with abuse potential tend to be more expensive on average.

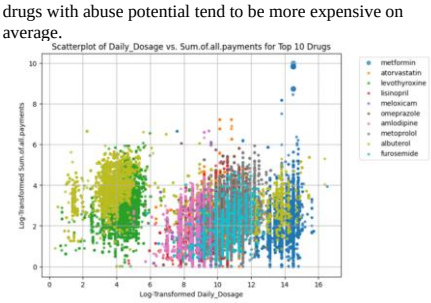


Figure 11. Scatterplot of sum of payment vs. daily dosage.

The relationship between daily dosage and payment sum is considered for ten select drugs (**Figure 11**). For Atorvastatin (orange), dosage is moderate and consistent, while the payment varies a lot. For Amlodipine (pink), the clustering is tight, indicating the dosage and payment both are consistent. For furosemide (cyan), there is a significant variation in both, as dosages and payments vary based on patient conditions and drug forms. For levothyroxine (green), there is high variability in both dosage and payment. For Lisinopril (red), the dosage seems to be consistent while payment is variable, probably due to brand cost difference. For Meloxicam (purple), there is significant variation in both dosage and payment. For Metformin (blue), dosage and payment appear to be standardized, indicating standardized prescribing and cost patterns. For Metoprolol (brown), dosage remains the about the same while payment sum is more variable. For Omeprazole (gray), there is a wide variation in dosage as well as payment, which might be due to OTC availability and different prescription forms. Overall, drugs like metformin and amlodipine have a tight clustering, which makes sense as they are common drugs that likely have a well-defined standard of care. Other drugs, like omeprazole, furosemide, and albuterol, have more dispersed points, which indicates that these medications are used in a wider variety of situations. Drugs such as atorvastatin, lisinopril, and metoprolol shows a relatively consistent dosing, which makes sense since they are used to treat chronic conditions. In contrast, the payments for these drugs show significant variability, which might be due to fact that for some, the expense is covered by insurance, leading to a variation in the out-of-pocket expense. Different demographic features in the data influenced dispersion in the Concentration variable (**Figure 12**). For all categorical features (e.g., sex, race, etc.), there are many outliers above the mean and IQR (**Figure 12a, 12b**). All IQRs are relatively short and have a lot of spread. There is no significant difference between categories in terms of dispersion. The concentration of drugs associated with income shows a trend reflective of income levels in society, with many lower-income points and a shrinking effect with many fewer points at high incomes (**Figure 12c**). Concentration does not have a clear relationship with age (**Figure 12d**).

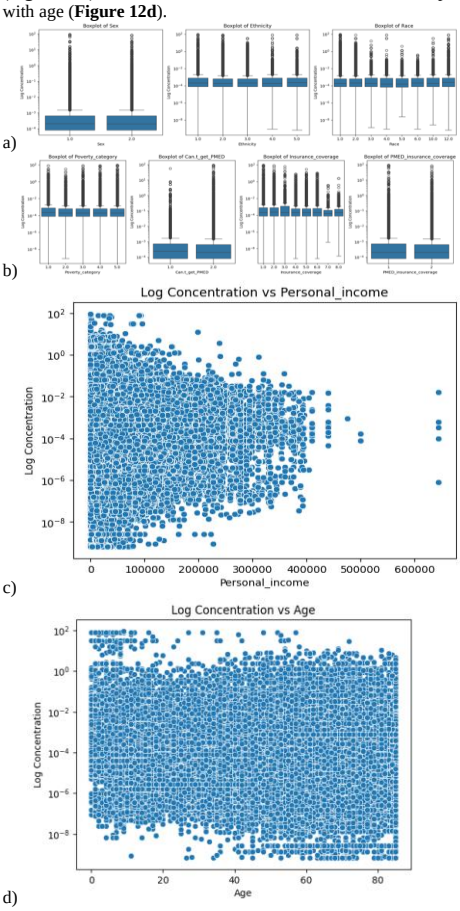
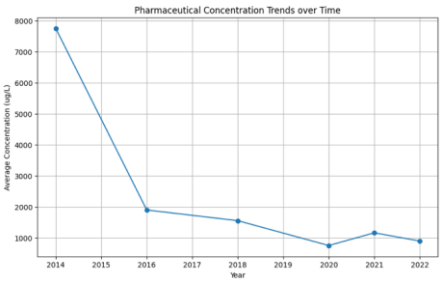


Figure 12. (a) Boxplot of sex, ethnicity, race against concentration; (b) boxplot of poverty category (poor to wealthy), availability of prescription medication (PMED), insurance coverage, and PMED insurance coverage against concentration; (c) concentration vs personal income; (d) concentration vs. Age.

Collecting several years of data afforded the opportunity to examine trends over time (**Figure 13**). Generally, pharmaceutical concentrations decreased over time between 2014 and 2022, with a much greater concentration in 2014 compared with the other years (**Figure 14a**). There was a modest increase in concentration between 2020 and 2021. To investigate if this increase was due to the onset of the COVID-19 pandemic, the yearly trends in total daily dosage of select cold and flu drugs (ibuprofen, dextromethorphan, hydroxychloroquine, acetaminophen, and diphenhydramine) were examined (**Figure 14b**). The dosage of these drugs was highest in 2014 and decreased until 2020, when they increased again slightly. However, it is unclear if this modest increase is due to the onset of the pandemic because dosage decreased between 2020 and 2021 and increased again in 2022 more significantly this time. Taken together, the overall effect of COVID-19 on dosage of cold and flu drugs entering the wastewater stream is unclear. The trends for antimicrobial and opioid drugs were also examined to investigate how the crises of antimicrobial resistance and opioid addiction have affected their wastewater concentrations over time (**Figure 14c, 14d**). Due to overall higher pharmaceutical concentrations in 2014, this year had the peak antibiotic and opioid concentrations. Both also experienced a peak in use in 2018, which is more pronounced for the antibiotics plot. Antimicrobial concentrations level off between 2020 and 2022, while opioid concentrations vary between modestly increasing and decreasing in those years. Overall, both antimicrobial and opioid concentrations have generally decreased since 2014. The stabilization of antimicrobial concentrations in more recent years suggests that doctors may be prescribing antimicrobials only when necessary in response to the uptick in antimicrobial-resistant infections globally. Due to increased variability, the prescribing trend for opioids is less clear, but the overall decrease in opioid concentrations suggests that doctors may be limiting opioid prescriptions.



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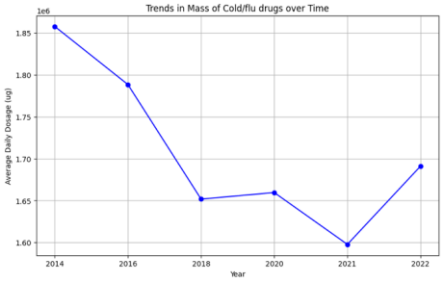

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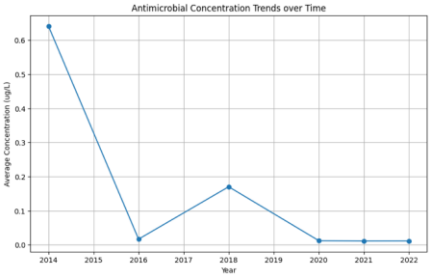

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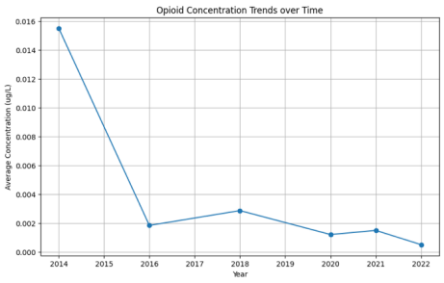


d)

Figure 13. Temporal trends between 2014 and 2022 for a) total pharmaceutical concentration, b) daily dosage of select cold and flu drugs, c) concentration of select antimicrobial drugs, and d) concentration of select opioids.

3.2 Model Implementation, Evaluation, and Insights

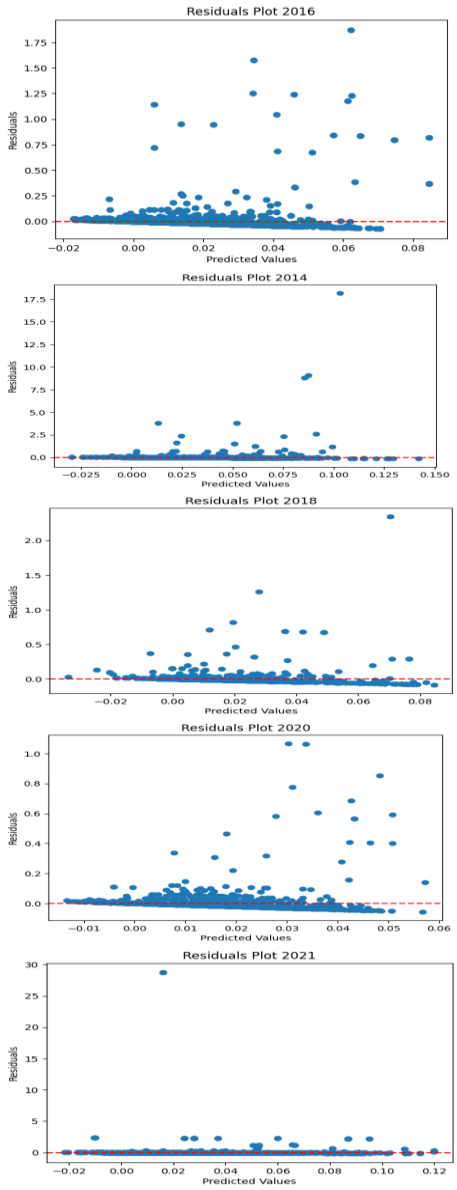
Six models were developed to predict the concentration profiles of different communities and predict temporal trends. These include linear regression, logisitic regression, multi-output k-nearest neighbor with principal component analysis, random forest classification, multi-output decision tree, and autoregressive integrated moving average models.

3.2.1 Model 1: Linear Regression The linear regression model was designed to predict the continuous variable Concentration using features that included demographic and socioeconomic variables as well as the categorical variable Drug. To ensure data quality, the dataset was preprocessed to remove missing or infinite values, and the features were standardized using StandardScaler. The dataset was then split into training (80%) and testing (20%) subsets to evaluate the model's predictive performance. The model was trained using the LinearRegression class from Python's scikit-learn library. Predictions for the test set were evaluated against the actual Concentration values, and several metrics were computed to measure the model's performance. Model accuracy was assessed using mean squared error (MSE), mean absolute error (MAE), R^2, and root mean squared error (RMSE), and a residuals plot was generated to visualize the differences between actual and predicted values (**Table 2, Figure 14**).

Table 2. Linear regression model evaluation metrics for each year.

Year	MSE	MAE	R-Squared	RMSE
2014	0.2760	0.0671	0.0084	0.5253
2016	0.0123	0.034	0.0310	0.1107
2018	0.011	0.0324	0.0096	0.1048
2020	0.0045	0.0254	0.01797	0.0674
2021	0.4711	0.0667	-0.0007	0.6864
2022	0.0088	0.0353	-0.0298	0.0938

From **Table 2**, MSE was lowest in 2020 (0.0045) and highest in 2021 (0.4711), indicating significant variation in prediction accuracy. MAE remained low overall, with the smallest value in 2020 (0.0254), showing better performance that year. The model explained little to no variance, with the highest R^2 in 2016 (0.0310) and negative values in 2021 (-0.0007) and 2022 (-0.0298). Finally, the RMSE improved from 2014 (0.5253) to 2020 (0.0674), although it increased sharply in 2021 (0.6864).



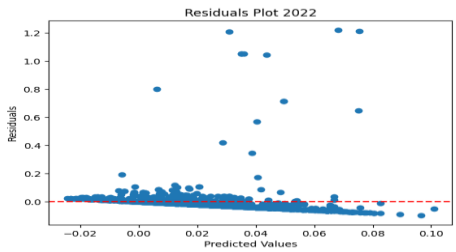


Figure 14. Residual plots for each year comparing residuals and predicted values.

From **Figure 14**, 2014 and 2021 have few outliers but show extreme residual magnitudes (e.g., above 17.5 in 2014 and 30 in 2021), which could heavily influence model performance. In contrast, 2016 and 2020 exhibit more frequent outliers, highlighting potential inconsistencies in the model's predictions. The residuals for 2016, 2018, 2021, and 2022 also fall below the red zero line, indicating instances where the model overpredicted the actual outcomes. Notably, in 2016, 2018, 2020, and 2022, heteroscedasticity is evident, as the spread of residuals increases with higher predicted values.

To examine the effect of each feature on model performance, each feature was removed (with the other remaining intact), and the resulting model was evaluated (**Table 3**). All reduced models show similar results, with minimal variations in the metrics. The MSE remains consistently around 0.1306, and the MAE ranges from 0.0422 to 0.0436, indicating that the model's prediction errors are relatively stable regardless of which feature is excluded. The RMSE stays around 0.2647 to 0.2651, suggesting that the model's predictions are consistently off by about 0.26 units of concentration. The R-squared values, which measure the proportion of variance explained by the model, are very low, ranging from 0.0038 to 0.0075, indicating that the model explains very little of the variability in concentration. Notably, the removal of the "Drug" feature results in a slight decrease in performance, with a negative R-squared value of -0.0006, suggesting that this feature may have been important to the model, as its removal worsened the fit.

Table 3. Linear regression model evaluation metrics with a given feature removed.

Feature Removed	MSE	MAE	RMSE	R2
Age	0.130807	0.043553	0.264979	0.003766
Sex	0.130617	0.043415	0.264738	0.006173
Ethnicity	0.130638	0.043434	0.264803	0.005134
Race	0.130609	0.043518	0.264790	0.004886
Personal_income	0.130600	0.043467	0.264704	0.006786
Family_income	0.130605	0.043408	0.264708	0.006830
Poverty_category	0.130612	0.043121	0.264702	0.007452
Cant_afford_PMED	0.130609	0.043487	0.264755	0.005714
Insurance_coverage	0.130645	0.043373	0.264769	0.005953
PMED_insurance_coverage	0.130611	0.043447	0.264732	0.006415
Drug	0.130733	0.042227	0.265103	-0.000567

Overall, the model's performance is poor, with low and negative R-Squared values. The model shows little improvement with the removal of any feature, indicating that better predictors or a different modeling approach may be needed to predict concentration more effectively. The model performed best in 2020, despite retaining overall poor performance that year as well.

3.2.2 Model 2: Logistic Regression The logistic regression model was developed to classify the categorical target variable Drug based on input features including Age, Sex, Ethnicity, Race, and various socioeconomic indicators like Personal income and Family income. To prepare the data, the target variable was label encoded, and missing or infinite values were handled by replacing them with NaN and dropping incomplete rows. The data was standardized using StandardScaler to ensure consistency across features, and the dataset was split into 80% training and 20% testing subsets using train_test_split from the scikit-learn library.

The model was trained using the LogisticRegression class from scikit-learn with the lbfgs solver and class_weight='balanced' to address potential class imbalances. Training was performed with a maximum of 2000 iterations to ensure convergence. The model's performance was evaluated using a variety of metrics, including accuracy, precision, recall, and F1-score (**Table 4**). Furthermore, the confusion matrix provided detailed insights into classification errors, highlighting areas where the model could be improved (**Figure 15**).

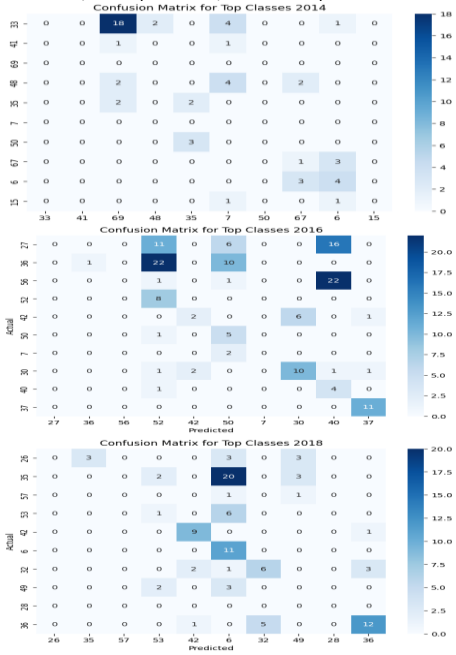
Table 4. Logistic regression model evaluation metrics for each year.

Year	Accuracy	Precision	Recall	F1-Score
2014	0.0482	0.0626	0.0482	0.0374
2016	0.0734	0.1384	0.0734	0.0633
2018	0.1061	0.1222	0.1061	0.0874
2020	0.0760	0.0937	0.0760	0.0619
2021	0.0639	0.2183	0.0639	0.0658
2022	0.0783	0.1902	0.0783	0.0734

From **Table 4**, the model performance was consistently low across all metrics. Accuracy ranged from 0.0482 (2014) to 0.1061 (2018), and precision, recall, and F1-score reached maximum values of just 0.2183 (2021), 0.1061 (2018), and 0.0874 (2018), respectively.

The confusion matrices for the years 2014, 2016, 2018, 2020, 2021, and 2022 illustrate the performance of the logistic model

over time (**Figure 15**). In 2014, the model showed moderate performance, with low counts across most classes and scattered misclassifications. By 2016, classification accuracy improved for certain classes, as indicated by stronger diagonal elements. The 2018 matrix demonstrated further improvement with concentrated diagonal values and fewer misclassifications. However, in 2020, the model's performance declined, with weaker diagonal elements and more scattered errors. The 2021 matrix showed notable improvement, with high accuracy for specific classes, though some misclassifications persisted. In 2022, performance appeared more generalized across classes, with evenly distributed misclassifications but weaker overall accuracy compared to 2021. These trends suggest varying performance across years, potentially due to changes in data distribution, model parameters, or feature importance.



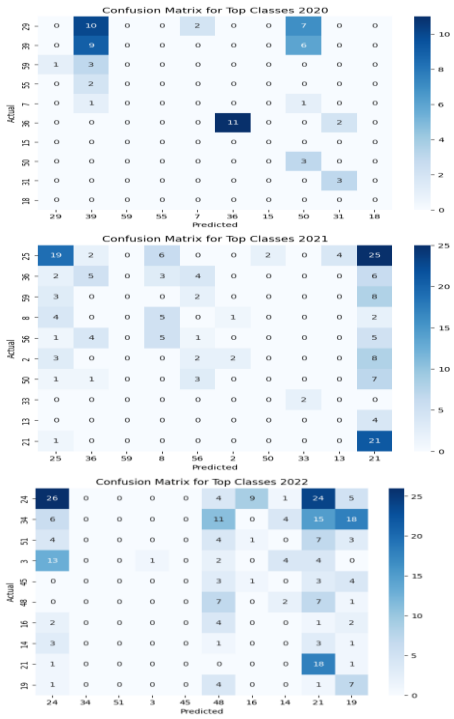


Figure 15. Confusion matrices for select drugs for each year. Each matrix compares the actual and predicted values for each drug.

The effect of each feature on model performance was additionally investigated in **Table 5**. Overall performance for all reduced models is low across all metrics. Accuracy remains under 0.1 for all reduced models, indicating poor model performance in distinguishing between classes. Both precision and recall are similarly low, suggesting the model struggles with identifying true positives while making many false positive predictions. The F1-score, which balances precision and recall, follows the same trend, being low across the board. Notably, the "Concentration" feature shows the worst performance, with accuracy of 0.038, precision of 0.081, recall of 0.038, and F1-score of 0.027, highlighting its importance to the model. On the other hand, the Can't Afford PMED feature produces the highest performance metrics, with an accuracy of 0.073, precision of 0.174, recall of 0.073, and an F1-score of 0.064, suggesting it has a relatively less impact on model performance. The Concentration feature appears to be the most critical, and removing it results in a noticeable decline in performance.

Table 5. Logistic regression model evaluation metrics with a given feature removed.

Feature Removed	Accuracy	Precision	Recall	F1-score

Age	0.063946	0.130498	0.063946	0.057531
Sex	0.064443	0.144547	0.064443	0.056868
Ethnicity	0.070503	0.148455	0.070503	0.061134
Race	0.070433	0.148597	0.070433	0.059946
Personal_income	0.066770	0.123383	0.066770	0.056435
Family_income	0.070704	0.160076	0.070704	0.062338
Poverty_category	0.066365	0.131556	0.066365	0.058656
Cant_afford_PMED	0.073155	0.174791	0.073155	0.064868
Insurance_coverage	0.068261	0.158311	0.068261	0.058547
PMED_insurance_coverage	0.069883	0.150317	0.069883	0.060637
Concentration	0.038291	0.081522	0.038291	0.027595

Overall, the logistic regression model struggled to perform well across all metrics and years. Low accuracy and F1-Scores indicate limited predictive power, while the consistently low precision and recall highlight challenges in identifying true positive instances. The year 2018 showed slightly better performance across all metrics, though the improvement was marginal.

The low performance across all reduced models suggests that the model may require better features, more data, or a different approach for improved accuracy in predicting the correct drug.

3.2.3 Model 3: Multi-Output KNN-PCA Model The multi-output k-nearest neighbor (KNN) model simultaneously predicted drug name and concentration in a single model based on features like income, insurance coverage, and demographics for each year. Prior to beginning the modeling step, attributes with too many missing values were excluded. Key feature engineering steps were also undertaken. Categorical features such as Sex and Ethnicity and the categorical target Drug were encoded into numerical features using LabelEncoder, and StandardScaler was employed to standardize the numerical features (since KNN models are sensitive to scale). Using train_test_split, the dataset was split into a training set (80% split) and a testing set (20% split).

Considering the characteristics of the dataset, which includes few similar features, rare events, and noise, Principal Component Analysis (PCA) was applied to reduce the feature space to just two components while preserving the most significant variance. This approach helped simplify the model, reduce computational costs, and maintain its predictive power. Notably, the reduction was done only on the datasets which showed less accuracy due to the presence of similar features (such as Personal Income/Family Income and Ethnicity/Race) and noise (i.e., from 2018 through 2020).

The model was constructed using scikit-learn's KNeighborsRegressor and MultiOutputRegressor together, which predicted drug name and concentration in parallel. The model was fine-tuned on the n_neighbors hyperparameter; the bias-variance tradeoff was most effectively combatted for values between 5 and 9. After running the model on the test set, model evaluation was performed (**Table 6**). For the concentration prediction, the MSE and R^2 score were mostly low but varied between years. For the drug prediction, the precision, accuracy, recall, and F1 score remained high for all years.

A confusion matrix (**Figure 16**) comparing predicted and actual values for 20 select drugs further confirms the accuracy of the drug predictions. Some common drugs such as Levothyroxine, Metformin, and Lisinopril are more likely to be correctly predicted, with fewer misclassifications, while others, such as Simvastatin, have some off-diagonal elements. In these cases, the model occasionally mistakes them for drugs like Hydrochlorothiazide and Amlodipine. This implies that the model is more accurate for frequently appearing drugs but struggles with less common or similarly labeled drugs. **Figure 17** further compares actual and predicted drug names. Some drugs (like Metformin) show good agreement between actual and predicted values, while others vary more widely. Comparing the frequency of actual and predicted drug names (**Figure 18**) shows similarly mixed results. For instance, drugs like Amlodipine and Simvastatin show very close alignment between the predicted and actual frequencies, showing the model can handle commonly encountered drugs quite accurately. However, drugs like Hydrochlorothiazide and Levothyroxine show discrepancies, with predicted frequencies much higher than the actual ones, possibly reflecting an overestimation by the model for these drugs. This shows that, for rarer drugs, the model tends to overpredict, possibly due to the class imbalance or insufficient representation of rare drug classes in the dataset.

Table 6. KNN-PCA model evaluation metrics and insights for all years.

Table 6. KNN-PCA model evaluation metrics and insights for all years.

Year	Classification Metrics (Drug)			Regression Metrics (Concentration)		PCA Applied	Insights
	Accuracy (%)	F1 Score	Recall	MSE	R^2 Score		
2014	99.32	0.9966	0.9999	1.0988	0.1349	No	Excellent drug classification, but poor regression performance.
2016	99.38	0.9969	0.999	0.0315	0.1436	No	Improved regression results; classification remained strong.
2018	99.36	0.9968	0.999	0.4334	0.1957	Yes	PCA simplified high-dimensional data, enhancing model performance.
2020	99.64	0.9982	1.0	0.0053	0.0247	Yes	PCA reduced noise; notable improvement in regression and strong classification.
2021	99.59	0.9979	1.0	0.0290	0.0639	Yes	PCA stabilized predictions; classification remained reliable.
2022	99.60	0.9980	1.0	0.0184	0.0971	Yes	Achieved best regression results with PCA; classification come out to be pretty good.

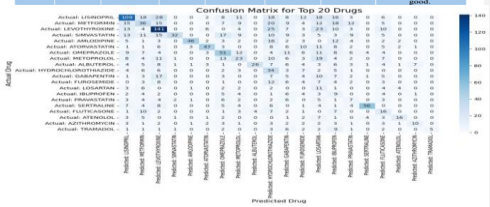


Figure 16. Confusion matrix comparing predicted and actual drug names for 20 select drugs.

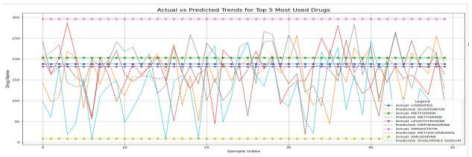


Figure 17. Line plot comparing actual and predicted drug names for select drugs.

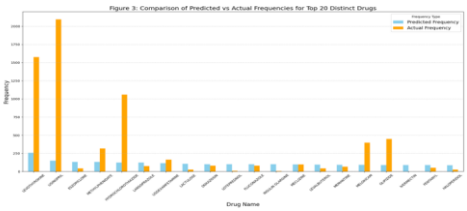
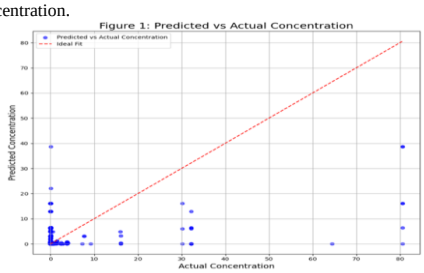
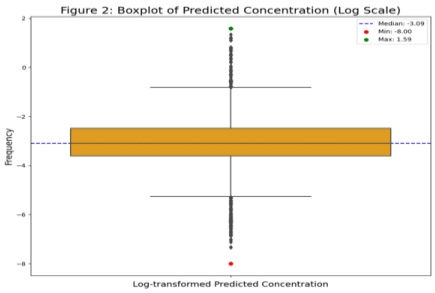


Figure 18. Bar plot comparing actual and predicted drug frequencies.

While the drug predictions have some mixed but generally accurate results, the concentration predictions are shown to perform poorly when visualized as well (**Figure 19**). There is a weak correlation between the predicted values and the actual values (correlation coefficient = 0.38) (**Figure 19a**). Many predictions diverge from the ideal fit represented by the red dotted line, notably at higher concentrations. There are many prominent outliers clearly indicating the model is not working well for Concentration. The median predicted log concentration is around -3, so half of the predictions fall below this value (**Figure 19b**). The log predicted concentration ranges from -4 and –2, meaning that most predicted concentrations are low. The minimum value is –8, indicating outliers, while the maximum value is 1.59. This shows that the model occasionally predicts concentrations that are much different from the concentrations of the majority of the pharmaceuticals. The whiskers range from -2 to –5, showing that the model tends to underpredict concentration.



a)



b)

Figure 19. a) Scatterplot comparing actual and predicted concentration values, and b) Boxplot of the dispersion of the concentration variable.

After fitting the full model, the impact of removing individual features from the model was explored as with Models 1 and 2. When the demographic features (e.g., age, sex, etc.) were reduced, the MSE for the concentration prediction increased from 1.0988 to 1.1457 and the R^2 score also dropped from 0.1349 to 0.0980, meaning the model made larger prediction errors and explained less of the variance in the data with fewer features. These changes suggest that the model relied on all the demographic features (especially age and ethnicity) to make accurate concentration predictions, although the range of predicted values for concentration after reducing features still showed a wide variation, with MSE ranging from 0.00001 to 1.5. Notably, for most years, when similar variables (e.g., Race and Ethnicity) were removed, there was a noticeable improvement in the MSE and R^2 score, suggesting that the model performs more effectively with an independent set of features.

For the drug predictions, the accuracy, F1 score, and recall remained consistent at 99.32%, 0.9966, and 0.9999, respectively, even when fewer features were used. The confusion matrix showed minimal changes, with a slight increase in false negatives (from 4 to 5) but overall, the classification performance remained very stable. The prediction of drug names remained the same mostly, like for high-frequency drugs such as propafenone and lovastatin being correctly identified. This means that demographics did not much affect the common drug usage, and the classification of drugs relies more on other features.

Overall, the predictions for drug names suggest that the classification results are trustworthy, but there is room for enhancement in the regression task for predicting concentration.

3.2.4 Model 4: Random Forest To predict drug name through another model, a random forest classification model was implemented using scikit-learn RandomForestClassifier. The regression was conducted using the fully integrated data for all years. Some minor data cleaning was required to correct for infinite values that were missed during the data cleaning step. Since this dataset is very large (1 million plus rows), a random sample of the data was taken of size 100,000. The features of interest were year, age, sex, race, ethnicity, personal and family

income, PMED access, and overall and PMED insurance coverage. Prior to running the model, the drug variable was label encoded. The train_test_split function from sklearn was used to split the data into a training set (80% of the data) and a test set (20% of the data). RandomForestClassifier was implemented with both default settings and with hyperparameter set {random_state=42, n_estimators=300, max_depth=10, max_features='log2', min_samples_split=10}. Both sets of hyperparameters were also used to implement the model with a stratified K-fold cross-validation approach, but the best performing model remained to be the default RandomForestClassifier with no modifications, which achieved 26% accuracy (**Table 7**).

Table 7. Random forest classification report for all features, reported from weighted average.

Accuracy	Precision	Recall	F1 Score
0.26	0.22	0.26	0.23

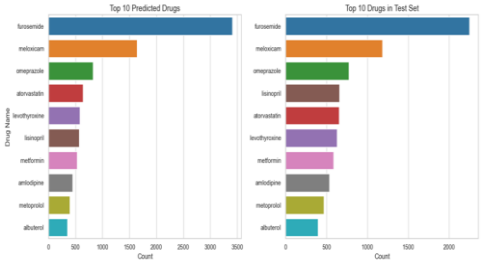
To see if a more accurate model could be achieved by feature reduction, the random forest was also subjected to individual feature reduction. The model accuracy, precision, recall, and F1 score increased substantially after applying each feature reduction (**Table 8**).

Table 8. Random forest classification report for different combinations of features. Each row has 1 feature removed and all others kept. Reported values are based on the weighted average. Note that SES is socioeconomic status.

Feature Removed	Accuracy	Precision	Recall	F1 Score
Age	0.60	0.61	0.60	0.59
Sex	0.62	0.62	0.62	0.61
Ethnicity	0.62	0.62	0.62	0.61
Race	0.62	0.62	0.62	0.61
Income/SES	0.32	0.35	0.32	0.29
Insurance coverage	0.62	0.62	0.62	0.61

Although each reduced model has similar performance, the best model performance was for the models that removed race and ethnicity, since they had the highest recall at 0.62 and greater or identical performance with other reduced models across the remaining metrics. Notably, recall is the most important performance metric since false negatives are more harmful when it comes to pollution than false positives. Additional analysis was conducted on this top-performing model to better assess its performance. The ten most frequent drugs in the predicted results and test set are identical, although the order of the most frequent drugs shifts somewhat (**Figure 20a**). Furthermore, the prediction accuracy of this model was 70% for a random sample of ten predicted drugs and ten actual drugs with the same indices (**Figure 20b**). Both of these results suggest relatively high model performance for the feature-reduced model.

a)



b)

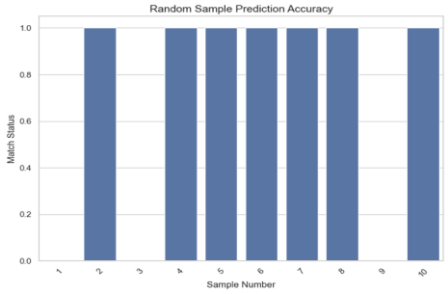


Figure 20. a) Comparison of the ten most frequent predicted drugs with the ten most frequent actual drugs, and b) prediction accuracy for a random sample of ten predicted and ten actual drugs.

Most likely, the similarities in performance for most of the reduced models are a sign of overfitting with the full model, since each of these features would be expected to affect model performance differently. For example, age is known to heavily influence the drugs consumed by a given individual, as incidence of heart disease, hypertension, and other diseases increases with age. In contrast, race and ethnicity are not as closely associated with particular pharmaceutical consumption relative to age. Furthermore, the model performs poorly when the socioeconomic variables are removed, suggesting these are important to the model performance. While the model without socioeconomic status still performs better than the full model, it seems these variables are more important than some of the other features. The importance of socioeconomic status is not surprising, since cost is a well-known barrier in the healthcare system. However, the relative importance of socioeconomic status with age, sex and other features still seems to be overstated by the model output when considered in the context of the medical literature. Overall, the model has promising results, but more refinement is needed.

3.2.5 Model 5: Multi output Decision Tree The multi-output decision tree model was used to predict drug name and concentration level (notably, low, medium, or high concentration; a categorical variable) in parallel using features such as age, sex, race, ethnicity, PMED access, insurance coverage, and medication attributes as predictors. The target variables, Drug and Concentration Level, were label-encoded to

numerical values for compatibility with the decision tree algorithm. Following data preparation, the dataset was split into training and testing sets using scikit-learn's train_test_split function using an 80% training and 20% test split.

The multi-output decision tree classifier achieved an accuracy of 95% for predicting Drug and 100% for Concentration Level (**Table 9**). Feature reduction was attempted as with Models 1-5, but the model exhibited identical performance for all reduced models. These results highlight the model's capability to handle multi-output classification effectively, with high precision in predicting both target variables. However, they also suggest that none of the features are too closely tied to predicting the drug concentrations, which is known to be untrue in the medical literature. Therefore, there is a need for further model validation and refinement to ensure robustness when applied to unseen data. This analysis demonstrates the utility of decision tree models for multi-output tasks in healthcare and socioeconomic studies.

Table 9. Multi-output decision tree model evaluation metrics for all years (combined).

PREDICTION	ACCURACY	PRECISION	RECALL	F1- SCORE	SUPPORT
Evaluation of Drug	0.95	0.95	0.89	0.95	319594
Evaluation for Concentration	1.00	0.88	0.87	0.95	125096

3.2.6 Model 6: ARIMA Forecasting future pharmaceutical concentrations into 2023 (for which data is not yet available) was attempted using the ARIMA (AutoRegressive Integrated Moving Average) model. While a powerful tool for time series analysis and forecasting, the ARIMA model struggled to deliver reliable results when applied to datasets with low observation counts. To implement the ARIMA model, the dataset was aggregated by drug and dosage across all years. After aggregation, the dataset consisted of approximately 400 records, with each drug typically having a maximum of six observations over time. This limited temporal depth per drug significantly hindered ARIMA's performance.

This is because an ARIMA model relies on historical patterns within the data, such as trends, seasonality, and noise, to make forecasts. With only six observations per drug, the model lacks the data density needed to identify and model these temporal patterns accurately. Furthermore, hyperparameter estimation for the ARIMA model on a small dataset can become highly unstable and cause overfitting. In addition to the size of the dataset, another challenge is in the high variability across individual drug records. For ARIMA to work effectively, the dataset must exhibit stationarity, where statistical properties like mean and variance remain constant over time. With fewer observations and high variability across records like in the current dataset, it becomes challenging to determine or transform the series into a stationary form, undermining the model's foundational assumptions. ARIMA additionally requires more lagged values to perform well, and the discrete values in this model do not allow it to forecast beyond immediately past trends. Because of all of these challenges, the ARIMA model could not be implemented successfully and demonstrated high error rates and low accuracy.

4 CONCLUSIONS

Overall, the best performing models were the KNN-PCA model for drug prediction (recall: 0.99) and the multi-output decision tree for concentration prediction (recall: 0.87). Most models were not sensitive to feature reduction, with the notable exception of the random forest model, whose accuracy increased 34 to 36% after removing each feature except those related to socioeconomic status. However, given the known relationships between features like age or race and pharmaceutical use, it is likely that the full random forest model was overfitting, which is why feature reduction generally improved performance. The random forest model performs poorly in the absence of the socioeconomic status variables, which suggests that these variables heavily influence pharmaceutical wastewater concentrations. Given that other features like age should have had a similar impact upon removal, it is unclear from the current analysis which features will be important in predicting pharmaceutical concentration with the highest recall, especially since the other models perform similarly before and after feature reduction. Notably, all models need refinement. No model performed well to predict the continuous concentration variable, although the multi-output decision tree can effectively predict low, medium, and high concentration levels, which may be informative to wastewater utilities in predicting the pharmaceuticals most likely to pass through wastewater treatment. To improve the prediction for concentration, a hyperparameter randomized search will be conducted for each model using scikit-learn's RandomizedSearchCV function. This was attempted during the course of the project, but ultimately was not successful due to insufficient computing power. Moving forward, these analyses will be run on a supercomputing node for improved results. Another possibility would be using the high-performing drug prediction models to predict the pharmaceuticals consumed in a given community and then link the predictions to the consumption data provided in the dataset. Finally, additional validation steps will be required before making a report to wastewater utilities. Specifically, the predictions will need to be validated against reported values in the literature or measured values in an experiment. These steps will be carried out moving forward with this project. Once the predictions are fully validated, they will be integrated into the SeweRx web application and shared with wastewater utilities across the United States. Utility operators will be able to use the predictions to cut monitoring costs, optimize pharmaceutical removal, and more effectively manage risk from pharmaceuticals with high concentrations or with known harmful traits like high ecotoxicity.

In addition to refining the concentration predictions from community characteristics, future models will include a temporal component to see how trends in pharmaceutical use will change over time. From the exploratory data analysis conducted in this study, pharmaceutical concentrations, including antibiotic and opioid concentrations, generally decreased between 2014 and 2022, and the mass of select cold and flu drugs did not increase much at the onset of the COVID-19 pandemic. This suggests that the crises of antimicrobial resistance and the opioid epidemic may be reducing prescription of antimicrobial and opioid drugs, and that the use of prescription cold and flu drugs is not impacted much by pandemics like COVID-19. However, future analyses will incorporate OTC drugs and other pharmaceutical traits, and

these may show an uptick in OTC cold and flu drugs in response to COVID-19 rather than prescription cold and flu drugs. Overall, these preliminary results suggest that there will be noticeable shifts in pharmaceutical wastewater concentrations in response to broad social and scientific changes like these public health crises. Notably, more work is needed to establish a stronger connection between these social shifts and shifts in pharmaceutical wastewater concentrations. Monitoring change over time or in response to these events will be informative for wastewater utilities in planning future projects and preparing for anticipated community needs during public health emergencies.

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